

Lesson 9.1: Reduction/simplification

Game theory

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Iterated dominance reduction/elimination

Example (4×4 game)

How will this two-player zero-sum game play out?

	b_1	b_2	b_3	b_4
a_1	0	1	7	6
a_2	4	2	3	4
a_3	3	1	0	2
a_4	0	0	7	3

- ▶ Round 1
- ▶ Round 2
- ▶ Round 3
- ▶ Round 4

Solution

Unique solution: (a_2, b_2) ; value 2 (favours row player)

Reduced sub-games

	b ₁	b ₂	b ₃	b ₄
a ₁	0	1	7	6
a ₂	4	2	3	4
a ₃	3	1	0	2
a ₄	0	0	7	3

► round 1

	b ₁	b ₂	b ₃
a ₁	0	1	7
a ₂	4	2	3

► round 2

	b ₁	b ₂
a ₁	0	1
a ₂	4	2

► round 3

	b ₁	b ₂
a ₂	4	2

► round 4

	b ₂
a ₂	2

Solutions: existence and uniqueness

► not all games reducible to unique solution/outcome:

	b ₁	b ₂
a ₁	2	0
a ₂	1	3

- game above doesn't have a unique *value*
- what can we say about such games?